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Simulation and optimization of cold chain logistics system towards lower carbon emission: a state-of-the-art review

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Abstract

Cold chain logistics is an emerging carbon emission source. The transportation section is estimated to generate more than 80% of its total carbon emission. Under the global warming background, a number of simulation and optimization studies have been thus conducted. Lately, cold chain logistics has been greatly aided by the quick advancements in data science and computer science. Consequently, the state-of-the-art developments in cold chain logistics system simulation and optimization were critically assessed in this research. The crucial indicators including optimization models, simulation models, and optimization algorithms were also quantitatively analyzed. It was discovered that, in single-objective optimization, carbon emissions were frequently included in the total cost objective function; but, in multi-objective optimization, they were utilized as a separate objective function. There were many studies on path optimization models, but site selection optimization models started relatively late and had high economic sensitivity, which might be a future trend. The ant colony algorithm was widely used in solving path optimization models. Genetic algorithms played a leading role in optimization algorithms, and they had different improvements and applications under different carbon reduction methods. Under the influence of multiple objective functions and constraints, multi-objective optimization algorithms might be a promising solution. The difficulties in modeling and refining the cold chain logistics system are introduced by the model accuracy, energy supply system revolution, carbon emission calibration, optimization result realization, and application scenarios that still exist. The outlook for the future is presented in terms of both technology orientation and demand orientation.

Highlights

- Carbon emission is emergingly used in both single- and multi-objective optimization, with the former usually included in total cost.
- Model development of site selection is more crucial than path planning due to its higher sensitivity to the total cost.
- Genetic algorithms dominate to optimize cold chain logistics systems, but multi-objective optimization algorithms show promise.

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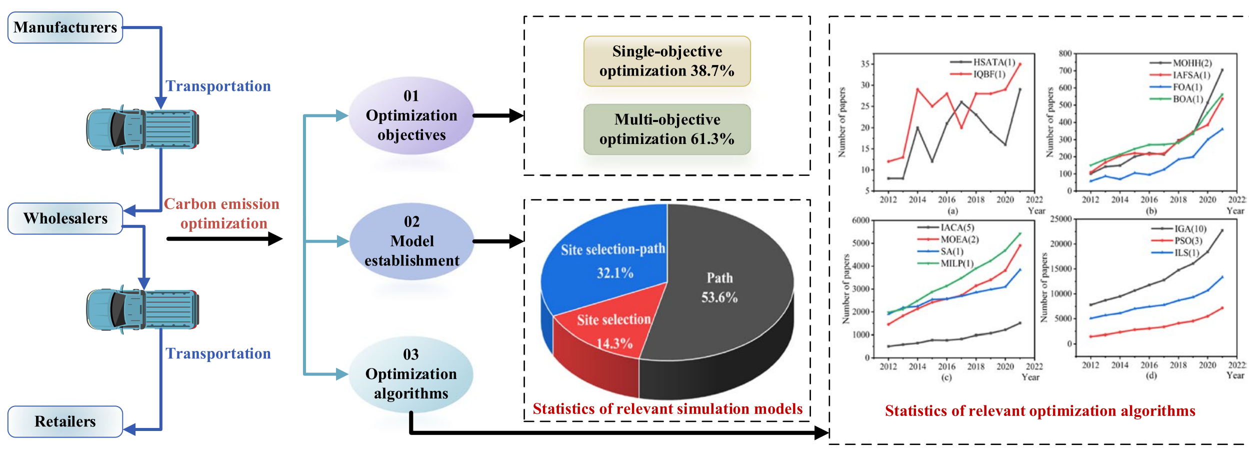
Keywords Cold chain logistics, Carbon emission, Simulation, Optimization, Path planning, Site selection

Graphical Abstract



1 Introduction

The logistics industry occupies a significant position in promoting social economy and living quality and strongly assists the development of other industries (Liu et al. 2021). In recent years, the world has witnessed a growing global demand for cold chain logistics products, such as fresh products (Liu et al. 2021; Ren et al. 2021; Zhang et al. 2020), pharmaceutical stuff (Nadimuthu et al. 2022; Wan et al. 2020; Zhang et al. 2021b), and processed food (Brito et al. 2012; Jedermann et al. 2014; Xu et al. 2017). Compared with traditional logistics, cold chain logistics systems are equipped with multiple refrigeration units to ensure the temperature required for product transportation, which also results in more carbon emissions. It was estimated that in 2010, the carbon emissions of the cold chain logistics industry accounted for almost 1% of the global carbon emissions (James et al. 2010). In 2020, the global cold chain logistics market was $233.8 billion, with an expected compound annual growth rate of 7.8% from 2021 to 2025 (Ba et al. 2021), indicating that cold chain logistics' carbon emissions are likewise rising dramatically. Consequently, there is a growing global interest in cold chain logistics system optimization as a means of lowering carbon emissions.

The total greenhouse gas emissions directly and indirectly were called carbon footprints generated by cold chain logistics activities (Hu et al. 2021). The life cycle assessment method was one of the most popular techniques for estimating carbon footprints (Dong et al. 2020). This evaluation method consists of four steps: (1) inventory analysis, (2) impact assessment, (3) target analysis and scope determination, and (4) result interpretation (Liu et al. 2022). The system boundaries for fruits and vegetables, for instance, comprise producers, distributors, merchants, consumers, and the abandoned. The typical system boundary for calculating carbon emissions in cold chain logistics systems is shown in Fig. 1a. Producers' carbon emissions were primarily seen during production, precooling, and refrigerated transportation from producers to distributors. The carbon emission of wholesalers was primarily manifested in storage, refrigerant leakage, and refrigerated transportation from wholesalers to retailers. The carbon emission of retailers was primarily manifested in storage, refrigerant leakage, and refrigerated transportation from retailers to showcases. The carbon emission of consumers was primarily manifested in storage (electricity consumption). When retailers deliver goods to consumers' homes, they often use insulated bags to maintain temperature and use electric vehicles for delivery, with almost no carbon emissions during this process. This article focused on carbon emissions during transportation, so we overlooked this situation when defining boundary conditions. The carbon emission of waste articles was primarily manifested in the rate of depletion (generally using landfill disposal). According to the carbon emission model of the cold chain logistics lifecycle, Hu et al. (2021) found that the carbon emissions of refrigerated transportation were much higher than those of other links. The study by Liu et al. (2022) found that 82% of the carbon footprint generated during the cold chain logistics transportation of fruits and vegetables could be traced back to the transportation process. Therefore, optimizing

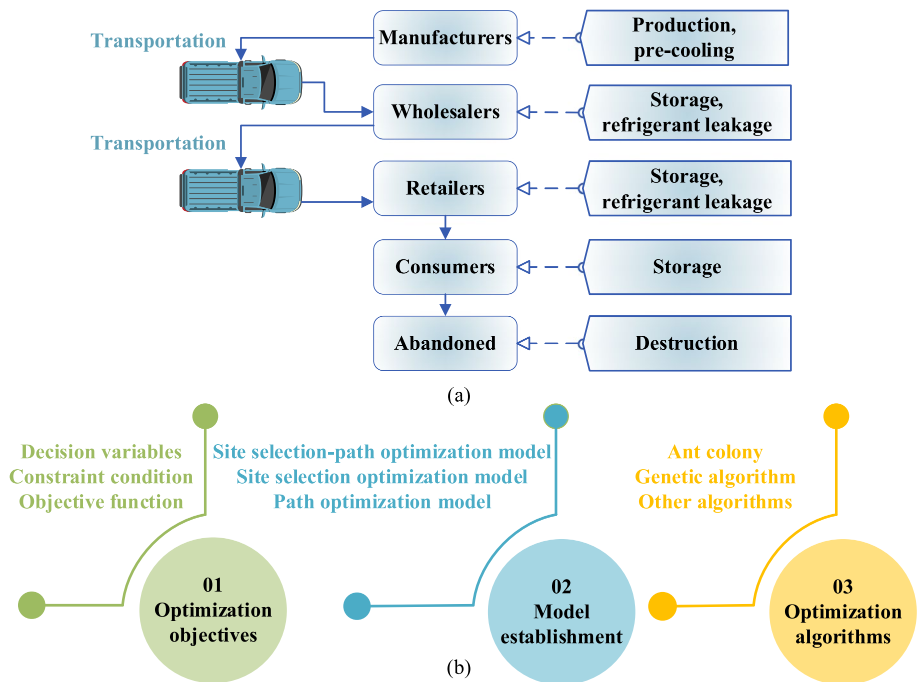


Fig. 1 (**a**) Common system boundary for carbon emission evaluation and (**b**) standard optimization procedure of cold chain logistics

the carbon emissions during the transportation process of cold chain logistics systems is particularly important.

As seen in Fig. 1b, the optimization procedure for cold chain logistics transportation systems involves defining optimization objectives, developing simulation models, and executing optimization algorithms (Ba et al. 2021; Chen 2020; Liu 2021; Zhang et al. 2003). The optimization objectives include decision variables, constraint conditions, and objective functions. Certain study beginning points, such as the economic perspective, environmental perspective, social perspective, and economic environment perspective, can be identified by identifying the optimization objectives of cold chain logistics. The model establishment can simulate the real-time cold chain logistics system in an economically efficient way, which makes the optimization of the path, site, and site selection path of cold chain logistics under complex circumstances simpler and cheaper (Carberry et al. 2014; Czocher 2017; Humphreys 2002). An optimization algorithm is a series of designed steps to scientifically obtain the appropriate parameters of the system. There were already many studies that simulate and optimize the cold chain logistics transportation process. Zheng Liu (Liu et al. 2021) and Bo Shu et al. (Shu et al. 2021) optimized the transportation of cold chain logistics for fresh products with the goal of minimizing the total cost. Lian Jie (2021) established a cold chain logistics optimization model under time window constraints, while Xinzhong Jia (2022) established a cold chain logistics distribution path optimization model under vehicle load and time constraints, reducing total costs. Yu Xiaosheng (2019) simulated and optimized the position of ship routes in cold chain logistics distribution, successfully reducing the total cost. Mo et al. (2022) established a multi-objective function to optimize the cold chain logistics of fresh agricultural products in Hainan and improve market competitiveness. However, the climate change concerns and recent progress in computer science/data science have led to some significant changes to the optimization objectives, simulation models, and optimization algorithms of cold chain logistics systems. On the one hand, compared with the traditional optimization objectives, modern optimization studies are now focusing more on the reduction of carbon emissions besides economic performance. On the other hand, numerous new modeling approaches and optimization algorithms have evolved as a result of the rapid development of machine learning. Therefore, it is urgently needed to comprehensively review relevant state-of-the-art advances, thus to constructively guide the research directions in this field.

According to the above background, this article extensively reviewed and analyzed the research on carbon emission optimization in cold chain logistics processes from the perspectives of optimization objectives, model establishment, and optimization algorithms and explores the opportunities and challenges for future development.

This article aimed to outline the carbon reduction methods of cold chain logistics, providing a breakthrough foundation for subsequent scientific and policy research.

2 Literature inventory for quantitative analysis

Quantitative analysis was conducted on relevant literature to obtain more objective results. The principle of collecting relevant literature and data was crucial for ensuring their representativeness when conducting quantitative analysis. The literature used for quantitative analysis was collected from Web of Science and ScienceDirect databases, and the keywords for the literature search were "carbon emission" and "cold chain logistics". Subsequently, a thorough search was conducted for specific terms such as target, model, and algorithm. After a comprehensive analysis of the content of the paper, only 30 papers were retained due to unsatisfactory relevance to the research topic. Figure 2 shows the keyword density visualization of these papers. This figure not only reflected the relevance of the literature but also indicated that models, optimizations, algorithms, carbon emissions, etc., were important for cold chain logistics. Quantitative analysis was conducted on the optimization objectives, simulation models, and optimization algorithms involved in this manuscript using VOSviewer 1.6.18 software.

3 Optimization objectives setups for cold chain logistics optimization

3.1 Cost indexes establishment

Cold chain logistics cost calculations include, but are not limited to, fixed costs, transit expenses, damage costs, refrigeration costs, time window penalty costs, and carbon emission costs. The symbols used in the following formula were explained in Table 1.

(1) Fixed costs, such as vehicle scheduling fees, vehicle depreciation, fixed maintenance costs, and labor costs, are unaffected by vehicle distance (Zhang et al. 2014a). The sum of all fixed expenses is represented by:

$$C_{Fixed\ cost} = \sum_{k=1}^{n} f_k s_k \tag{1}$$

(2) Transportation costs include fuel consumption during vehicle delivery and vehicle maintenance costs,

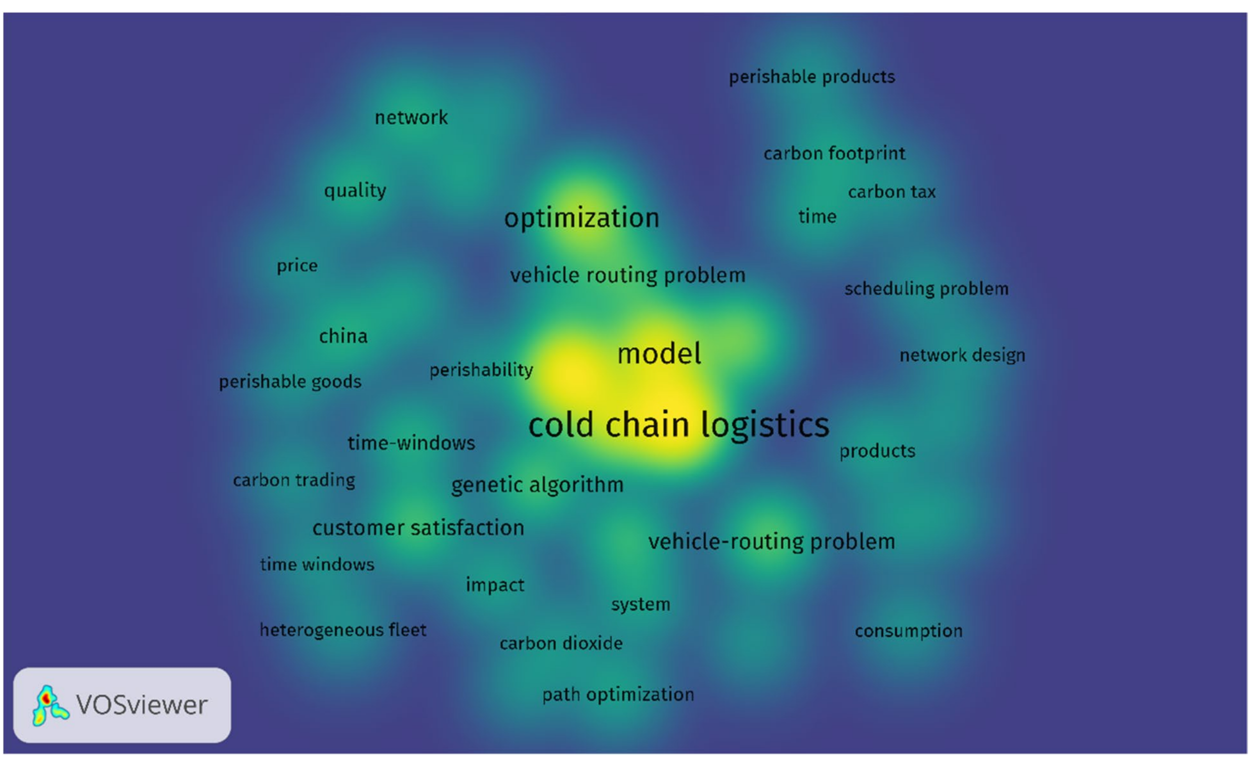


Fig. 2 Density visualization of keywords in relevant papers

Table 1 Description of symbols

Symbols	Description
f_k	The fixed cost of vehicle k
s_k	0–1 variable indicates whether the vehicle k has departed. If it has departed, $s_k = 1$; otherwise, it = 0
C_{ij}^k	The 0 – 1 variable represents the cost of transporting the vehicle k from i to j. When vehicle k needs to be transported from i to j, x_{ij}^k to 1, otherwise x_{ij}^k to 0
Y	The price of the unit product
x_{ij}^k	0–1 variable, the customer in j needs the k^th vehicle, $X_j^k = 1$, and vice versa, $X_j^k = 0$
θ_1	The mass loss rate per time unit while moving a vehicle
θ_2	The rate at which the object loses weight per unit of time upon coming into contact with the external environment
T_j^k	The moment that car K reaches the j^th demand point
T_i^k	The moment that car K departs from the i^th demand point
t_j^k	The time when the item comes into contact with external hot air
Q_j^k	The vehicle's remaining mass when it gets to j
α	The vehicle k's hourly energy consumption rate from i to j
β	The vehicle k's hourly energy consumption rate while unloading at site j
ε_1	The hourly cost of cars coming earlier than the customer's chosen time frame
ε_2	The hourly cost of cars coming later than the customer's chosen time frame
E_j	The earliest time of arrival that the client is willing to accept
L_j	The latest time of arrival that the client is willing to accept
φ_0	The amount of fuel used per unit of travel while the car is empty
φ^*	The amount of fuel used per unit of travel when the car is completely loaded
φ	The vehicle's maximum load capacity
φ_{ij}	The freight volume of vehicles from i to j
c_0	The price per carbon emission unit
η	The factor of carbon emissions

usually proportional to distance (Wang et al. 2021). The cost of transportation can be stated as follows:

$$C_{Transportation\ cost} = \sum_{k=1}^{n}\sum_{i=0}^{m}\sum_{j=0}^{m} C_{ij}^k x_{ij}^k \tag{2}$$

(3) The two main components of the cost of damage are the following: the first is the cost of quality damage resulting from time spent in transit; the other is the cost of quality damage resulting from contact with hot air from the outside, such as food arrival, code inspection (Wang et al. 2021).

$$C_{Damage\ cost} = Y\sum_{k=1}^{n}\sum_{j=1}^{m} x_{ij}^k[\theta_1(T_j^k - T_i^k) + \theta_2 t_j^k]Q_j^k \tag{3}$$

(4) The overall expense of refrigeration is divided into two components: the cost of keeping the cargo cold during transportation and the cost of keeping it cold after unloading (Chao et al. 2019; Wang et al. 2021). Refrigeration costs can be computed using the amount of refrigerant utilized. The temperature outside, the temperature the car must maintain, the vehicle's surface area, the heat transfer coefficient, and other factors all affect how much refrigerant is used.

$$C_{Refrigeration\ cost} = \sum_{k=1}^{n}\sum_{j=1}^{m} x_{ij}^k[\alpha(T_j^k - T_i^k) + \beta t_j^k]Q_j^k \tag{4}$$

(5) Time penalty costs include potential product quality degradation during the waiting period if the vehicle arrives too early and potential disruption to normal sales if it arrives too late (Zhang et al. 2014b; Zhu et al. 2021). This cost arises when the vehicle fails to

deliver the product to the customer within the designated time window.

$$C_{Time\,penalty\,cost} = \varepsilon_1 \sum_{k=1}^{n} \sum_{j=1}^{m} max\{T_j^k - E_j, 0\} + \varepsilon_2 \sum_{k=1}^{n} \sum_{j=1}^{m} max\{L_j - T_j^k, 0\} \tag{5}$$

(6) In the distribution process of cold chain logistics, the carbon emission cost is equal to the carbon dioxide emission cost. This includes the energy and refrigerant usage carbon emission costs associated with cold chain logistics transportation, which are impacted by vehicle load and distance traveled.

The relationship between vehicle load, transportation distance, and fuel consumption per unit of distance is as follows (Zhang et al. 2014a):

$$E(\varphi_{ij}) = \frac{\varphi^* - \varphi_0}{\varphi} \varphi_{ij} + \varphi_0 \tag{6}$$

The following is the price of carbon emissions:

$$C_{Carbon\,emissions\,cost} = c_0 \eta E(\varphi_{ij}) x_{ij}^k \tag{7}$$

3.2 Optimization objectives setups for cold chain logistics optimization

Single objective optimization refers to an optimization problem that includes an objective function and possible constraints. Multi objective optimization involves two or more optimization problems, where multiple objective functions conflict with each other (Choo et al. 2023; Gujarat 2023). Table 2 shows the cost functions in relevant publications. According to the number of objective functions, the carbon emission optimization objectives of cold chain logistics systems can be divided into two categories: single objective optimization and multi-objective optimization. The classification of optimization objectives in published literature is shown in Fig. 3. 61.3% was single-objective function optimization, and 38.7% was multi-objective function optimization. Single-objective function optimization generally refers to minimizing the total cost. The total cost generally consisted of fixed costs, transportation costs, damage costs, refrigeration costs, time window penalty costs, and carbon emission costs. Sometimes it also included production costs, transformation costs, enterprise costs, government subsidy costs, etc. Multi-objective function optimization would simultaneously consider issues such as minimizing total cost, minimizing carbon emission cost, maximizing customer satisfaction, minimizing user cost, minimizing waiting time, and achieving optimal product quality.

Optimization objectives were divided into constrained optimization and unconstrained optimization (Zakaria et al. 2012). Finding a feasible solution to one or more mathematical functions in a constrained search space is called constrained optimization (Segura et al. 2016). As seen in Fig. 4, published studies were arranged in this review according to the three restrictions of vehicle load, time window, and customer demand. Currently, 16.7% of articles had not considered constraints during the research process. Vehicle load was the most common single constraint condition, accounting for 33.3%. The proportion of customer demands and time window constraints was relatively small, accounting for 3.3% and 6.7%, respectively. Simultaneously considering vehicle load and time window was a common dual constraint condition, accounting for 26.7%. There was also a dual constraint condition that considered both vehicle load and customer demand, accounting for 6.7%. There was relatively little research on the topic of considering vehicle load, customer demand, and time window constraints simultaneously, accounting for 6.7%. It could be seen that in any situation, vehicle load was a very important constraint condition.

3.2.1 Single-objective optimization

A single-objective function was used in single-objective optimization to solve issues. In single objective optimization, there were 15 articles studying total cost minimization, one article studying carbon emission cost minimization, and one article studying unit customer cost minimization. Among all single objective functions, minimizing total cost was the most common function type, accounting for 89.54%. The single objective functions of minimizing carbon emission costs and minimizing unit customer costs both accounted for 5.23%. The optimization objective of traditional cold chain logistics transportation procedures did not include carbon emission costs. This article did not design an objective function or select a model and algorithm to optimize carbon emissions in the cold chain logistics transportation process, but extensively reviewed and analyzed the research on carbon emission optimization in cold chain logistics processes from the perspectives of optimization objectives, model establishment, and optimization algorithms.

When considering carbon emission costs for optimization objectives, the total cost would increase, but the

Table 2 Summary of cost functions in relevant publications

Reference	MTC											MCC	MCS	MDP	MUC	PQO
	FC	TC	RC	PC	DC	CC	SC	IC	WC	WO	OP					
(Zhang et al. 2014a)	√	√									√	√				
(Zhang et al. 2014a)												√				
(Wang et al. 2018)	√	√	√	√	√	√										
(Wang et al. 2017)	√	√	√	√	√	√	√									
(Li et al. 2019)	√	√			√	√		√								
(Zhang et al. 2019a)	√	√	√	√	√	√										
(Ren et al. 2019)	√	√		√	√	√										
(Chen et al. 2019b)	√	√	√		√	√									√	
(Qin et al. 2019)	√	√	√		√	√							√			
(Chen et al. 2019a)	√	√		√		√										
(Wang et al. 2020b)	√	√	√		√							√	√			
(Liu et al. 2021)		√	√	√	√			√								
(Li et al. 2020)	√	√	√	√	√	√										
(Liu et al. 2020b)	√	√	√	√	√	√										
(Babagolzadeh et al. 2020)	√	√	√		√	√										
(Utama et al. 2020)	√	√				√										
(Li et al. 2021)									√	√		√	√			
(Zhang et al. 2021c)		√				√			√	√					√	
(Zhao et al. 2020)	√	√			√							√	√			
(Leng et al. 2020a)	√	√			√	√							√			
(Wang et al. 2020a)		√	√	√	√					√				√		
(Leng et al. 2020c)	√	√			√				√				√			
(Leng et al. 2020b)	√	√				√			√				√			
(Khan et al. 2020)	√	√		√		√										
(Deng et al. 2021)	√	√	√	√	√	√	√									
(Chen et al. 2021)	√	√	√	√	√	√										
(Zhang et al. 2021a)	√	√	√	√	√	√										
(Qu et al. 2021)	√	√		√		√										
(Ning et al. 2021)		√		√		√										
(Lian 2021)	√	√	√		√								√			
(Golestani et al. 2021)		√	√			√			√							√

cost of dealing with the impact of carbon emissions was greater than the increase in total cost. So considering the cost of carbon emissions was beneficial for both the economy and the environment (Hariga et al. 2017). Many articles had studied strategies such as carbon pricing, carbon quotas, and taxes to save carbon emission costs (Babagolzadeh et al. 2020; Chen et al. 2019a, 2021; Li et al. 2019; Liu et al. 2020b; Ning et al. 2021; Qin et al. 2019; Wang et al. 2018, 2017). The simultaneous action of cost and constraints could improve economic, environmental, and social benefits. Vehicle load, time window, and customer satisfaction were common constraints. Single-objective functions often only constrained vehicle loads. Many costs in the total cost were calculated under ideal conditions, without considering the impact of uncertainties such as vehicle failures, sudden weather changes, and customer importance on costs. We should conduct more thorough research on policies and find carbon emission regulations that were suitable for enterprises and governments according to local conditions.

3.2.2 Multi-objective optimization

Multi-objective optimization could optimize two or more objective functions simultaneously which was a vector optimization (Ozdemir et al. 2019; Pan et al. 2021). Figure 3 shows that in the optimization of the cold chain logistics transportation process, 76.7% of the models were bi-objective optimization, and 23.3% of the models had three objective functions. For multi-objective optimization, minimizing the total cost was a fixed objective

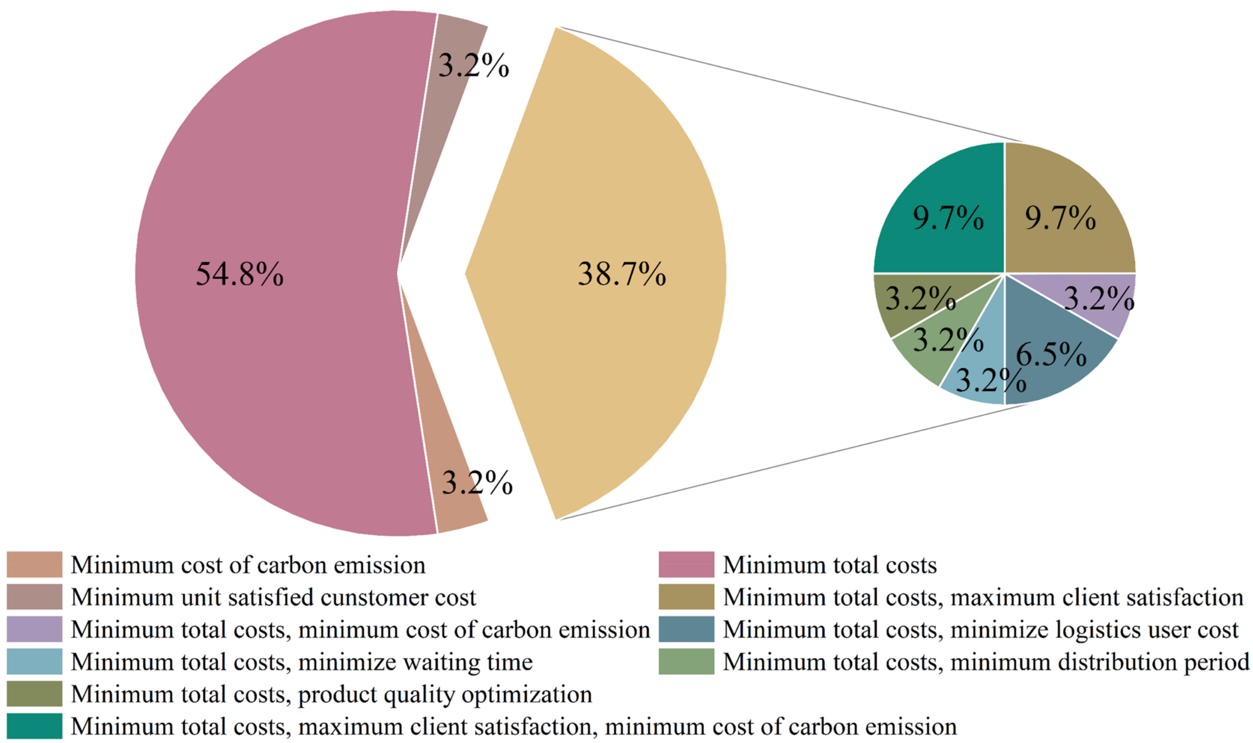


Fig. 3 Proportions of different optimization objectives

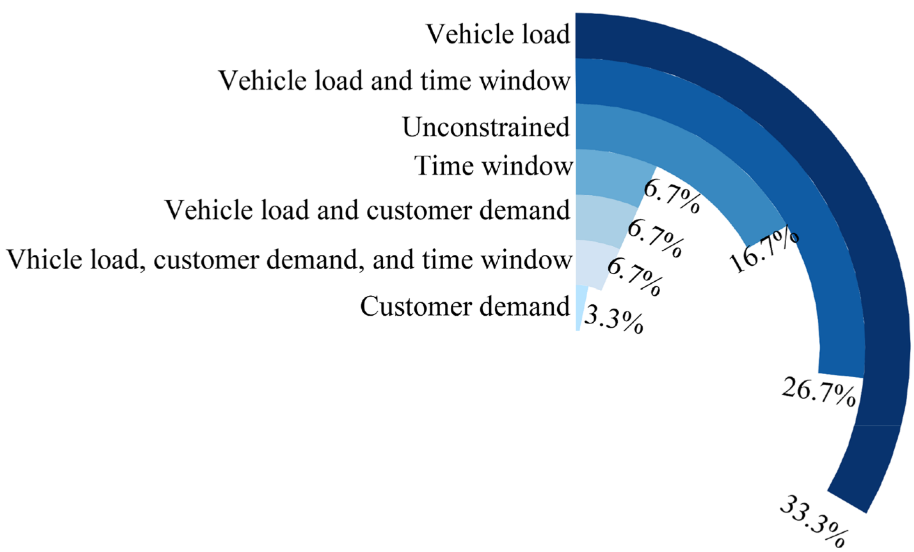


Fig. 4 Proportions of constraint conditions for carbon emission optimization

function. For path optimization models, consideration would also be given to minimizing carbon emission costs and maximizing customer satisfaction. For the site selection optimization model, consideration would also be given to minimizing logistics user costs, minimizing waiting times, minimizing delivery cycles, and achieving the best product quality. At present, site selection models mainly solve multi-objective problems. For the site selection-path planning model, customer satisfaction issues would also be considered. Compared to single-objective optimization, multi-objective optimization was more complex. The multi-objective solution of the low-carbon cold chain and supply chain was a complex, non-deterministic polynomial problem. There is currently no deterministic algorithm that could solve this problem. Finding an effective approximation algorithm was the direction of efforts to solve this problem. In single objective optimization, the cost of carbon emissions was usually included in the total cost. Whether it was single-objective optimization or multi-objective optimization, carbon emission cost was a basic component.

4 Simulation models supporting the prediction of cold chain logistics designs

Thanks to simulation models, virtual replicas of real cold chain logistics systems could be established at reasonable costs. As illustrated in Fig. 5, a keyword co-occurrence network was established for the searched literature, and it was found that the word model was a high-frequency word in the keywords. The model keyword co-occurrence network shows that modeling could address issues such as time windows, carbon emissions, and customer satisfaction. Therefore, model establishment was crucial to the article. The cold chain logistics' carbon emission optimization was modeled by 53.6% of the path, 14.3% of the site selection, and 32.1% of the site selection path.

4.1 Path planning models

The transportation method for cold chain logistics was optimized by taking into account both environmental and financial benefits. A vehicle path optimization model with the least total cost (including carbon emission cost) was created to encourage the growth of cold chain logistics toward a low-carbon economy (Chen et al. 2021, 2019b; Liu et al. 2021; Utama et al. 2020). Also, general path optimization models with a minimum total cost and carbon emission cost (Ning et al. 2021; Zhang et al. 2014a), multi-level path optimization models (Wang et al. 2020b), and heterogeneous fleet-vehicle path optimization models had also been established (Li et al. 2020). Cold chain logistics began to take into account small batches, time window, freshness, client satisfaction, and other issues to achieve better economic, environmental, and social benefits in the context of intense competition (Wang et al. 2018). This helped them become more competitive.

Thus, a joint delivery path optimization model (Liu et al. 2020b), a path optimization model taking into account the carbon tax policy (Wang et al. 2017), a multi-objective path optimization model taking into account customer satisfaction (Li et al. 2020; Qin et al. 2019), a

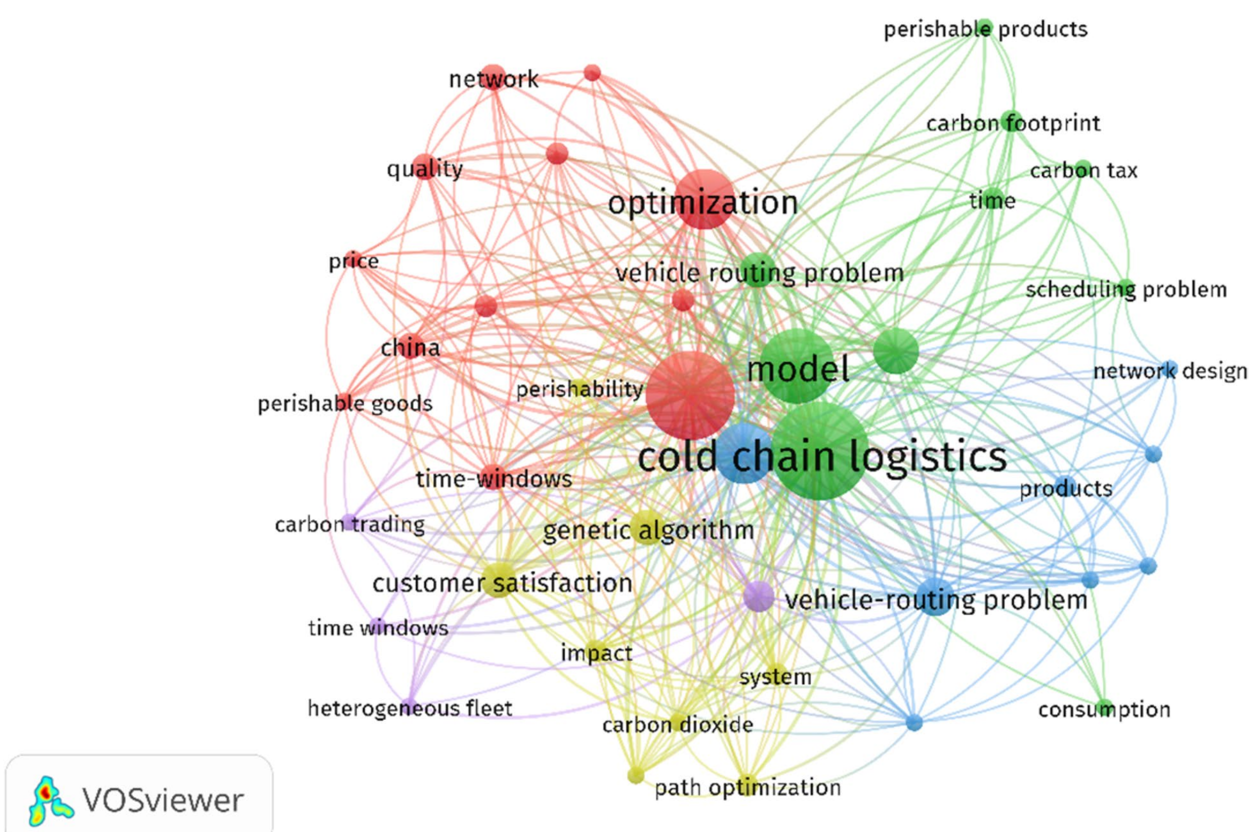


Fig. 5 Keyword symbiosis network for research on cold chain logistics carbon emission optimization

path optimization model with minimum delivery cost (Lian 2021), and a path optimization model taking into account food quality and total cost were established (Chen et al. 2019a). Both single- and multi-objective optimization models were taken into consideration in the study of cold chain logistics' transit phase. Nonetheless, the percentage of models for multi-objective optimization was comparatively low. While taking into account restrictions like small batch size, time window, freshness, and customer satisfaction, the path optimization model for cold chain logistics' transportation stage sought to lower economic costs and carbon emissions. However, the model was simulated under ideal conditions and did not take into account real-world factors.

4.2 Site selection models

Site selection in cold chain logistics would also cause carbon emissions. In the four articles collected so far for site selection optimization, one published in 2020 and three published in 2021, could be seen that the research on site selection optimization started relatively late. Securing economic, environmental, and social benefits while resolving multi-objective problems was the primary goal of site selection optimization. The site selection optimization model in the cold chain logistics transportation stage resolves the issues of maximizing product quality and reducing overall system costs (Golestani et al. 2021). Explored topics included minimum costs for construction and operation, minimum costs related to carbon emissions, and minimum costs related to customer satisfaction penalties (Li et al. 2021). Simultaneously, it addressed the issues of decreasing both user and overall costs, encompassing expenses related to damage, refrigeration, transportation, fuel consumption, time window penalties, and distribution center operations (Chen et al. 2019b).

Compared to path optimization models, site selection models were more sensitive to economic costs. Different objective functions were considered for the location model. However, there were currently few studies on location models, and there was no research on objective functions with constraints. The model was established under ideal conditions without considering real-world factors.

4.3 Site selection-path planning models

The significant carbon emissions linked to this industry have made it imperative to design the most cost-effective and efficient cold chain logistics (Leng et al. 2020a). To confirm the significance of researching carbon emissions in the cold chain logistics transportation process, a site selection path model was developed as early as 2017 (Wang et al. 2018). A cold chain logistics site selection path optimization model was developed in 2019 to examine the overall cost and carbon emissions under various carbon emission policies to identify more appropriate carbon emission policies (Li et al. 2019; Zhang et al. 2019b). Because of the perishability and sensitivity of cold chain items, by 2020, considerations would include product quality (Khan et al. 2020), client satisfaction (Leng et al. 2020c; Wang et al. 2020a), and other social impacts. In the transportation stage of cold chain logistics, site selection paths were optimized not only for total costs and carbon emissions but also by frequently taking social, environmental, and economic benefits into account at the same time. So far, the uncertainty factors in carbon emission policy and the transportation process were also the research focus of this model. With the deepening of research, the uncertainty factors such as carbon emission policy, client demand (Qu et al. 2021), and vehicle model (Babagolzadeh et al. 2020) would be taken into account simultaneously when building the model, bringing it closer to reality.

5 Optimization algorithms towards efficient and effective cold chain logistics systems

The optimization algorithm for the low-carbon cold chain logistics transportation stage was mainly heuristic. By conducting numerous trials, heuristic algorithms could produce workable solutions to difficult issues in a fair amount of time (Ozdemir et al. 2019). Heuristic algorithms were algorithms that were based on intuition or experience. When problem features were missing, actual data was limited, or unavailable and could not be used for complex problems, they were the best choice for making quick and wise decisions. Feasible solutions might not necessarily be the optimal solution (Al-Shaery et al. 2022). Simple heuristics, meta-heuristics, and hyper-heuristics were three types of heuristic algorithms. Numerous search and optimization issues could be resolved with the use of heuristic algorithms (Abdollahzadeh et al. 2021; Yang 2011).

The shipping process has been optimized using a variety of heuristic algorithms to lower carbon emissions from cold chain logistics. As illustrated in Fig. 6, 13 heuristic algorithms were now in use to optimize cold chain logistics' carbon emissions. Figure 6 annotates the articles that optimized carbon emissions in cold chain logistics and highlights the publications that used these techniques between 2012 and 2021. With around half of the total, the most widely used algorithms were the enhanced ant colony and improved genetic algorithms. In the meantime, 10% of research publications optimized carbon emissions during the cold chain logistics transportation stage using the particle swarm optimization algorithm. Furthermore, according to earlier studies, roughly 7% of researchers use

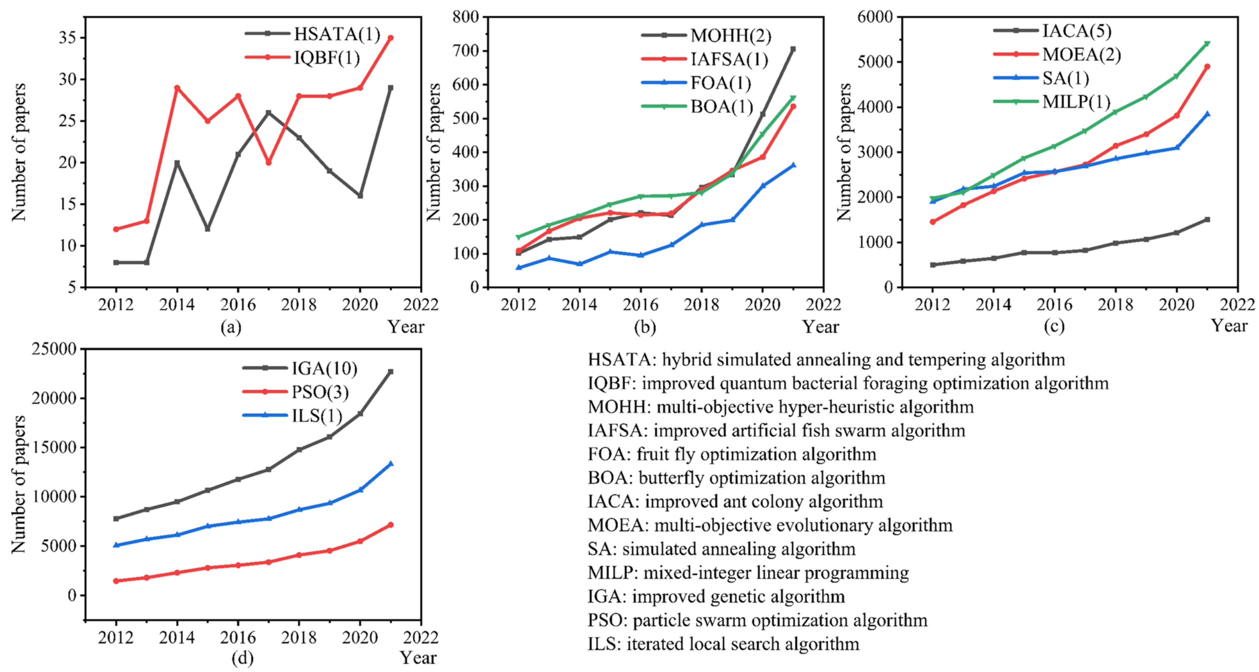


Fig. 6 Summary of the optimization algorithm (numbers in parentheses represent how many of the 30 articles in this review use this algorithm)

multi-objective hyperheuristic and multi-objective evolutionary algorithms to choose the best transportation optimization. Iterative local search algorithms, mixed integer linear programming, enhanced quantum bacterial foraging optimization algorithms, fruit fly optimization algorithms, hybrid simulated annealing and tempering algorithms, simulated annealing algorithms, and butterfly optimization algorithms are among the remaining algorithms. The carbon emission optimization challenge in the cold chain logistics transportation stage could be resolved with the use of these algorithms.

5.1 Improved ant colony algorithm

The ant colony algorithm was a method for determining optimization pathways that were inspired by natural (Blum 2005) ant colony behavior (Feng et al. 2007; Gajpal et al. 2006; Parpinelli et al. 2002; Zhang 2019). When ants look for food, they produce pheromones along the way they go, and other ants follow the trail they sense. Ant colony algorithms were characterized by strong robustness, positive feedback (Marcoulides et al. 2003), strong global search capability, easy combination with other methods (Zhang 2019), and slow convergence (Huang et al. 2021; Liu et al. 2017; Zeng et al. 2016).

Carbon emissions and overall cost were the two main areas of attention for cold chain logistics improvement. When determining the optimization model during the transportation phase, the ant colony algorithm might enter local optima (He et al. 2021; Huang et al. 2021; Kazharov et al. 2010). However, Zhang. et al. (2019) found that integrating the ant colony algorithm with the ribonucleic acid algorithm to calculate path optimization models increased global search capabilities and successfully lowered total costs and carbon emissions. One of the biggest problems in cold chain transportation was how to keep fresh food fresh. Chen et al. (2019a, b) combined the ant colony algorithm and the nucleic acid algorithm to develop a path optimization model that considered freshness. After the sensitivity of fuel and carbon costs were examined, it was discovered that freshness decreased with increasing fuel prices and freshness increased with increasing carbon prices. By analyzing the relationship between temperature and damage coefficient using the improved ant colony algorithm, Zhao et al. (2020) discovered that the greater the temperature difference, the greater the amount of carbon emissions caused, and the smaller the damage coefficient, the smaller the economic cost.

As research and technology advance, some real-world scenarios will be taken into account while optimizing low-carbon cold-chain logistics transportation. The artificial fish colony algorithm and the ant colony algorithm were coupled by Chen et al. (2019a, b) to develop pathways for smaller quantities of commodities with lower costs and lower carbon emissions. Client satisfaction was not immutable, and the introduction of a penalty factor according to different satisfaction-generated penalty costs could better meet customer needs (Zhao et al.

2020). The degree of air pollution in the transportation process is constantly changing. It was discovered that the more severe the pollution level, the more economical the method was when combining the DNA and ant colony algorithms to solve path optimization models under varied pollution levels (Zhang et al. 2014a).

The study solved a low-carbon cold chain logistics path optimization model by combining the ant colony technique with other algorithms. The influencing factors of total cost, carbon emission cost, and freshness cost were analyzed. It resolved the low-carbon cold chain logistics transportation optimization problem in real-world scenarios with low product quantities, high customer satisfaction, and varying air pollution levels. The ant colony algorithm itself did not have the advantage of solving the model. Combining it with other algorithms could help calculate the path optimization model. Improved ant colony algorithms could examine model sensitivity, lower transportation costs, and find solutions in a variety of real-world scenarios.

The parameters influencing the overall cost, freshness cost, and carbon emission cost were examined using the improved ant colony algorithm. A low-carbon cold chain logistics transportation stage optimization problem was solved under realistic circumstances, including small batch production, customer satisfaction, and varying air pollution levels. The ant colony algorithm itself did not have the advantage of solving the model. The improved ant colony algorithm could reduce transportation costs, analyze the sensitivity of the model, and solve problems in different actual situations. Investigating rational ways to integrate the ant colony algorithm with other algorithms was worthwhile. Currently, the combination of the ant colony algorithm with other algorithms was only used to solve the path optimization model for low-carbon cold chain logistics, but not for the site selection model and the site selection path model.

5.2 Improved genetic algorithm

Drawing on the principles of natural organism evolution, a genetic algorithm was suggested. Better optimization outcomes could be found more quickly when solving difficult issues by mimicking chromosomal selection, crossing, and mutation (Nearchou 1998). The genetic algorithm that covered a large area, which was conducive to solving nonlinear problems (Guo et al. 2020), was inherently easy to realize parallelization (Liu et al. 2020a), and had strong robustness. However, while tackling optimization problems in low-carbon cold chain logistics transportation operations, standard genetic algorithms might run into local optima (Chen et al. 2020) and slow convergence speeds (Andre et al. 2000; Wen et al. 2021). It was discovered that this genetic algorithm was simple to integrate with other algorithms and could enhance global search ability and convergence speed by comparing the optimization results of traditional genetic algorithms and the combination of genetic algorithms and simulated annealing algorithms (Li et al. 2020). It was discovered that optimization during the transportation phase was crucial after using an improved genetic algorithm to solve the carbon emission optimization model of cold chain logistics (Deng et al. 2021). It was discovered that including carbon emission cost would raise the overall cost but enhance the environmental benefits by incorporating it into the optimization process of the cold chain logistics transportation stage (Wang et al. 2018). Policies pertaining to carbon emissions and the amount of carbon emissions were closely correlated (Khan et al. 2020). Carbon emission regulations included carbon emission taxes (Huang et al. 2020; Toptal et al. 2014), carbon emission quotas and trading (Huang et al. 2020; Toptal et al. 2014; Yang et al. 2014), carbon emission ceilings and compensation (Li et al. 2019), and carbon emission ceilings (Huang et al. 2020; Toptal et al. 2014). When it came to carbon emission quotas and trade, the amount spent overall and the amount of carbon emissions would fluctuate depending on how adequate the carbon quotas were (Qin et al. 2019). However, the cost resulting from carbon trading regulations only made up a small part of the overall cost; instead, other criteria like customer satisfaction are frequently taken into account when determining carbon quotas and pricing (Wang et al. 2020b). Compared to other policies, the carbon tax mechanism had a minimum carbon emission but also a maximum total cost (Qin et al. 2019). When the carbon tax was too large or too small, the impact of changes in carbon tax on carbon emissions was minimal. Within a certain range, carbon emissions would decrease with the increase in carbon tax, indicating that a reasonable carbon tax was beneficial for reducing carbon emissions (Wang et al. 2017).

As research advances, the optimization method must be taken into account how to make the model more realistic in addition to expenses. Zhang et al. (2021a, b, c) solved a low-carbon cold chain logistics path optimization model with several distribution centers and demand locations by combining the simulated annealing approach with the genetic algorithm. Client satisfaction was a major challenge in the current situation and was not necessarily high when the total cost of distribution was low. Li et al. (2021) established the penalty cost resulting from low customer satisfaction to resolve this problem. They then solved the location model with the lowest total cost using a non-dominated sorting genetic algorithm.

To solve the path, location, and location path optimization models of low-carbon cold chain logistics, genetic algorithms were combined with other methods. The relationship between economic costs and environmental benefits under carbon tax and carbon quota trading policies was discovered, and the low-carbon cold chain logistics transportation stage optimization problem was resolved by taking into account multiple demand points, multiple distribution centers, and customer satisfaction. The improved genetic algorithm was utilized more frequently than alternative algorithms in the resolution of the transportation stage model for cold chain logistics. The price and volume of carbon emissions change depending on the carbon emission policy. Under the carbon trading policy, the transportation phase accounts for the least amount of carbon emission costs, but under the carbon tax policy, the transportation phase accounts for the least amount of carbon emissions. However, as domains like big data and machine learning continue to grow, how to combine genetic algorithms with other algorithms more effectively was a challenging problem. At the same time, more realistic factors should be considered in the solving process.

5.3 Other algorithms

Many more algorithms have also been used, in addition to the genetic and ant colony algorithms, to optimize the cold chain logistics transit stage model while taking carbon emissions into account. Many might believe that using the quickest route had the most impact when it came to low-carbon cold chain logistics optimization during the transportation stage. The shortest approach might not always have the lowest overall cost and carbon emissions, according to research on cold chain logistics path optimization utilizing an enhanced quantum bacterial foraging optimization algorithm (Ning et al. 2021). One of the main areas of interest in low-carbon cold chain logistics research was carbon emissions. Different policies affect the link between carbon emissions, carbon pricing, and total costs.

Cloud particle swarm optimization algorithm (Zhang et al. 2021c), chaotic particle swarm optimization algorithm (Zhang et al. 2019b), simulated annealing algorithm path (Liu et al. 2020b), driving method and robust optimization theory (Qu et al. 2021), hybrid simulated annealing and tempering algorithm (Chen et al. 2021), iterative local search algorithm and mathematical algorithm of mixed integer programming (Babagolzadeh et al. 2020) were employed in the investigation of the connections among carbon pricing, carbon emission, total cost, carbon quota, carbon subsidy, and carbon tax. Changes in carbon prices were found to affect total costs and carbon emissions (Oliveira et al. 2020). It was discovered that adopting carbon quota regulations could lower overall expenses and carbon emissions when compared to not adopting carbon quota rules. When carbon prices remain unchanged, the total costs and carbon quotas are negatively correlated. The relative magnitude of carbon emissions and carbon quotas determines the connection between carbon prices and overall costs when carbon emissions are constant. The total cost rises in tandem with rising carbon prices when carbon emissions are high, whereas the total cost falls in tandem with rising carbon prices when carbon quotas are high. The biggest influence on energy saving and emission reduction comes from the combination of carbon emission subsidies and carbon emission quota laws. Businesses would be more motivated to make better decisions as a result of carbon taxes, which would lower overall costs and carbon emissions. Carbon emissions could be decreased with a fair carbon price (Hariga et al. 2017).

In actuality, there were a lot of unknowns involved in low-carbon cold chain logistics transportation. By optimizing the model with the particle swarm optimization algorithm, it was found that considering uncertain factors would bring better economic, environmental, and social benefits (Chen et al. 2019b). The uncertain factors affecting carbon emissions were studied by using multi-objective evolutionary algorithms (Leng et al. 2020a), iterative local search algorithms, and mathematical algorithms of mixed integer programming (Babagolzadeh et al. 2020), fruit fly optimization algorithms (Lian 2021), hybrid simulated annealing and tempering algorithms (Chen et al. 2021), iterative local search algorithms and mathematical algorithms of mixed integer programming (Babagolzadeh et al. 2020), and hybrid butterfly optimization algorithms (Utama et al. 2020). Studies have indicated a strong correlation between carbon emissions and both vehicle load and travel distance. Shortening the path could greatly reduce carbon emissions. Deterioration of traffic conditions would increase carbon emissions. Carbon emissions were also impacted by the site selection center's capability. In the current situation, client satisfaction, time windows, heterogeneous vehicles, perishable cold chain items, and so on were all constraints. These practical factors have been taken into account in the optimization calculation of the cold chain logistics transportation stage by the fruit fly optimization algorithm (Lian 2021), the multi-objective evolutionary algorithm (Leng et al. 2020a), and the iterative local search algorithm (Babagolzadeh et al. 2020). The optimization results were more in line with the actual situation.

In conclusion, path, site selection, and site selection-path optimization models for low-carbon cold chain logistics could all be solved using these algorithms. The study found out why the optimization of the low-carbon cold chain logistics transportation stage did not study the shortest path alone; examined how carbon prices, overall expenses, emissions, carbon quotas, carbon subsidies, and carbon taxes relate to one another; identified some uncertain factors that affect carbon emissions, and analyzed the optimization of low-carbon cold chain logistics transportation under some practical constraints. These algorithms have not yet been widely applied to the optimization of cold chain logistics transportation processes and were only used to solve one problem in the optimization process. Multi-objective optimization algorithms had great potential as they could simultaneously solve both objective functions and constraints. The priority should be to find out whether an algorithm could be used to solve all the problems faced in the optimization process.

The path planning model is currently the most researched, mainly solving single-objective optimization problems. The objective function generally solved the problem of minimizing the total cost, including fixed costs, transportation costs, refrigeration costs, penalty costs, damage costs, and carbon emission costs. The path planning model optimized transportation paths under various constraints. The site selection model mainly solved the optimization of multi-objective functions. The objective function would consider construction costs, operating costs, carbon emission costs, transportation costs, customer satisfaction costs, etc. The most common constraint condition was the vehicle load. The site selection-path planning models optimize single-objective and multi-objective functions under the constraints of time window and vehicle load. When optimizing a single-objective function, carbon emission-related policies will be discussed. Under the constraints of vehicle load and time window, the variables of the objective function usually include costs in path planning models and site selection models. The three optimization models would consider the cost of carbon emissions. Simulations were conducted under relatively ideal conditions, with less consideration given to the influence of real-life factors. Compared with the path planning model, the site selection model was more sensitive to economics, and the site selection-path planning model was more sensitive to carbon emission-related policies. The improved ant colony algorithm was only suitable for solving path-planning models. Improved genetic algorithms could be used to solve path, site selection, and site selection-path planning models.

6 Future perspectives

6.1 Challenges of low-carbon cold chain logistics

Based on the results of the above review, it could be concluded that reducing carbon emissions through the optimization and simulation of cold chain logistics systems was a viable approach. However, challenges in aspects of model accuracy, energy-supplying system revolution, carbon emission calibration, optimization result realization, and application scenarios still existed.

(1) Model accuracy. Even though a number of cutting-edge models, particularly machine learning models had been applied in this field, it was hard for the developed models were hard to consider all the influential factors, some of which might significantly affect the model accuracies. For example, both weather and traffic conditions were important influential factors, but they were usually missing in simulation and optimization models due to their complexity. Besides, it was found that the most accurate models in this field are static. The development of accurate dynamic models demanded.

(2) Energy-supplying system revolution. When simulating the energy consumption and relevant cost in a cold chain logistics system, it was usually assumed that the traffic was powered by gasoline or diesel. In fact, because of the revolution of energy-supplying systems of vehicles, namely the emerging application of electric vehicles, the adaptability of traditional simulation and optimization models was required to be modified and updated.

(3) Carbon emission calibration. Carbon emission calibration has always been a difficult task in most industries. The cold chain logistics system was an especially hard one due to its large and ambiguous edge. The difficulty in calibration means it was hard for us to validate the accuracy of developed simulation and optimization models in the carbon emission aspect, leading to possible errors in the results.

(4) Optimization result realization. A pathetic fact was that, even if inspiring optimization results for low-carbon cold chain logistics systems had been obtained, the optimal design might not become a reality due to some subjective and objective factors, user preference, official approval of the selected location, unconcerned financial cost, etc. In many cases, the optimization results might offer merely a reference instead of an application design.

(5) Application scenarios. The concerned traffic approach in traditional simulation and optimization studies of cold chain logistics systems was usually road transportation. Modern life in cities showed an emerging demand for air and water transport. While relevant simulation and optimization studies were relatively limited.

6.2 Opportunities of low-carbon cold chain logistics

Although optimization and simulation of cold chain logistics systems faced the above technical challenges, there were also many opportunities in this field, which lie in technology-oriented and demand-oriented aspects.

(1) Development of artificial intelligence. New advances in cold chain logistics system simulation and optimization were possible with the growth of artificial intelligence. The functions of machine learning, ChatGPT, and other software were very powerful and had been widely used in some fields. Their functions were worth learning and could be migrated from other fields to the simulation and optimization of cold chain logistics.
(2) Development of the Internet of Things. The accuracy of models used in cold chain logistics systems has increased with the emergence of Internet of Things technologies. The advancement of tools and technology like satellites, GPS, and GIS allows for more precise measurement of carbon emissions during transit and efficient analysis of cold chain logistics' carbon footprint.
(3) Development of transportation technologies. Compared to gasoline and diesel, the price and cost of electricity were typically more stable. The revolution of electric vehicles could more or less enhance the robustness of the simulation and optimization models. On the other hand, the advancement of the Internet of Things and autonomous driving could help lessen the barriers to implementing optimization outcomes.
(4) Emerging demand for cold chain logistics. The cold chain logistics market grew at an average yearly pace of more than 20% between 2013 and 2017 (Li et al. 2021). In the cold chain logistics sector, scientific decision-making has grown in significance as the market grows, and increased standards have been set for simulation and optimization accuracy.
(5) Growing concern about the carbon emission problem. To protect the ecological environment of the earth, greenhouse gas emissions have become a global hot issue (Fan et al. 2022; Sun et al. 2022). To fulfill the aims and commitments of economic development and carbon reduction (Zhao et al.), optimizing carbon emissions in the logistics industry was an essential technique and method. As a result, a study into the high carbon emission logistics business was critical, since it provided chances for optimizing carbon emissions in cold chain logistics.

7 Conclusions

This review provided a comprehensive insight into the simulation and optimization of carbon emissions in cold chain logistics systems, including optimization objectives, simulation models, optimization algorithms, and the challenges and development opportunities faced by the simulation and optimization of cold chain logistics systems.

Single-objective optimization research was more popular in the carbon emission optimization stage of cold chain logistics transportation than multi-objective optimization. Both single-objective optimization and multi-objective optimization considered carbon emission costs. The path planning model mainly solved single-objective function optimization problems. Site selection models often solve multi-objective function optimization problems. The site selection-path planning model could optimize both single-objective and multi-objective functions. When the objective function of the site selection-path planning model was a single objective, the impact of carbon emission-related policies on path planning would be discussed. Optimizing the objective function ignored the impact of uncertain factors such as vehicle malfunction, sudden weather changes, and customer importance on costs when calculating costs. The carbon emission optimization modeling in the cold chain logistics transportation stage mainly focused on the path planning model. The site selection model was more sensitive to economic costs, while the site selection path planning model was more sensitive to carbon emission-related policies. In the model solving of the cold chain logistics transportation stage, the improved ant colony algorithm could only calculate the path planning model, while the improved genetic algorithm was widely used in path planning, site selection, and site selection-path planning model solving. The multi-objective optimization algorithms had great potential.

Simulation and optimization of the cold chain logistics system were promising approaches to reducing carbon emissions. However, challenges in aspects of model accuracy, energy-supplying system revolution, carbon emission calibration, optimization result realization, and application scenarios still existed. Although optimization and simulation of cold chain logistics systems were faced with challenges, there were also many opportunities in this field, which lie in technology-oriented and demand-oriented aspects. The development of artificial intelligence such as machine learning, ChatGPT, and other software has promoted the simulation and optimization of cold chain logistics systems. The development of the Internet of Things, represented by tools and technologies such as satellites, GPS, and GIS, can more accurately measure carbon emissions and improve the accuracy of cold chain logistics system models. The development of transportation technologies such as electric vehicles and autonomous driving can help reduce obstacles to achieving optimization results and enhance the robustness of simulation and optimization models. With the continuous growth of demand for cold chain logistics, the standards for simulating and optimizing carbon emissions models in cold chain logistics are becoming increasingly high. The increasing attention to carbon emissions has made optimizing carbon emissions in the logistics industry an important technology and method, providing opportunities for optimizing carbon emissions in cold chain logistics.

Abbreviations

MTC	Minimum total costs
FC	Fixed cost
TC	Transportation cost
RC	Refrigeration cost
PC	Penalty cost
DC	Damage cost
CC	Carbon emission cost
SC	Shortage cost
IC	Inventory holding cost
WC	Construction cost
WO	Operation cost
OP	Overload punishment cost
MCC	Minimum cost of carbon emission
MCS	Maximum client satisfaction
MDP	Minimum distribution period
MUC	Minimize logistics user cost
PQO	Product quality optimization
IACA	Improved ant colony algorithm
IGA	Improved genetic algorithm
PSO	Particle swarm optimization algorithm
MOEA	Multi-objective evolutionary algorithm
MOHH	Multi-objective hyper-heuristic algorithm
HSATA	Hybrid simulated annealing and tempering algorithm
SA	Simulated annealing algorithm
IAFSA	Improved artificial fish swarm algorithm
FOA	Fruit fly optimization algorithm
ILS	Iterated local search algorithm
MILP	Mixed-integer linear programming
IQBF	Improved quantum bacterial foraging optimization algorithm
BOA	Butterfly optimization algorithm

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Authors' contributions
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Data availability
The datasets used or analyzed during the current study are available from the corresponding author on reasonable requests.

Declarations

Competing interests
Guanyi Chen is an editorial board member of *Carbon Research* and was not involved in the editorial review, or the decision to publish, this article. All authors declare that there are no competing interests.

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